Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. The learning rate was set to 0.001 with the Adam optimizer.

2 Results

Figure 1 shows the accuracy of LSTM by dataset.

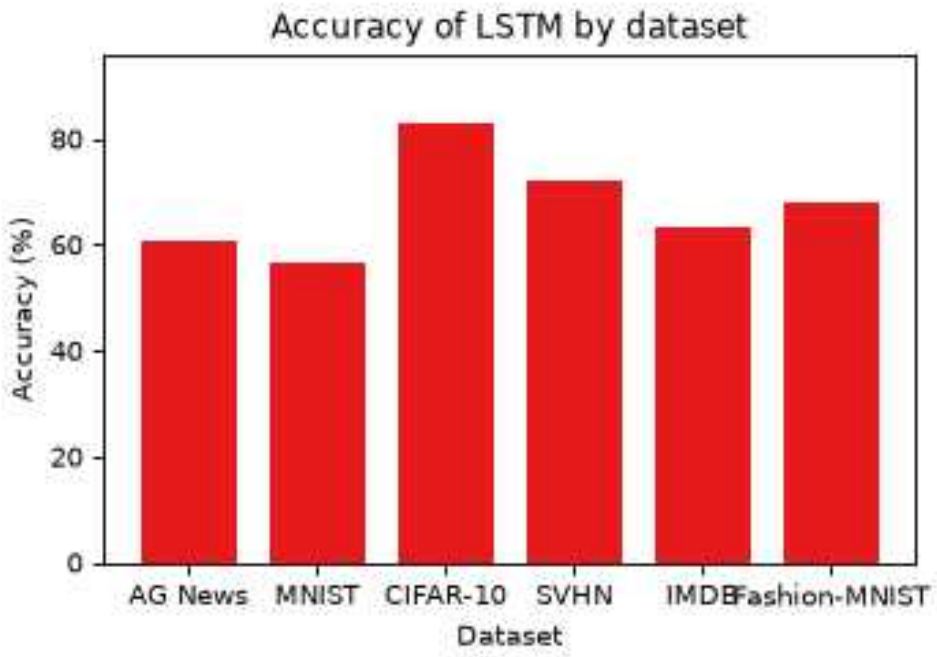


Figure 1: Accuracy of LSTM by dataset for the models in this experiment.

LSTM scores 34.3 points higher on CIFAR-10 than on Fashion-MNIST (Figure 1). LSTM scores 7 points higher on Fashion-MNIST than on AG News (Figure 1). LSTM scores about 4 points higher on AG News than on MNIST (Figure 1).

Figure 2 shows the f1 score during training.

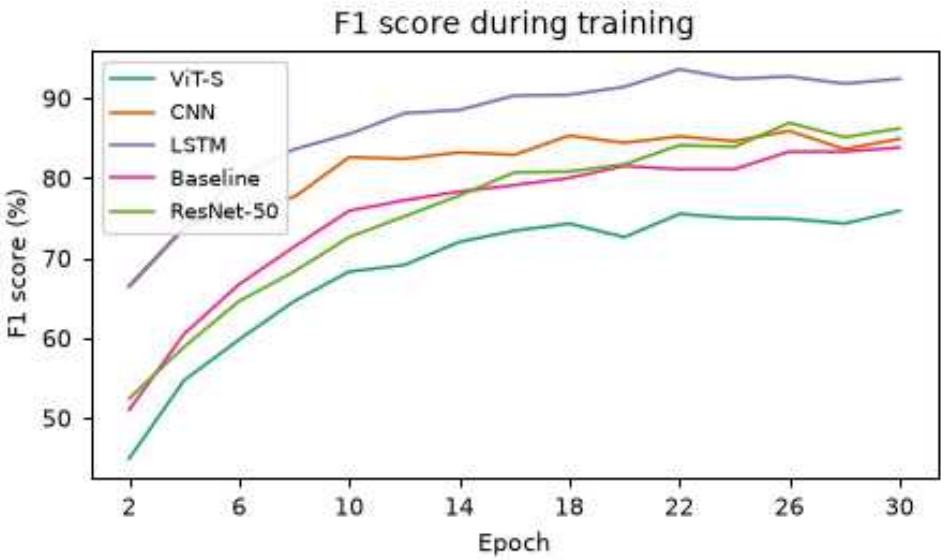


Figure 2: F1 score during training for the models in this experiment.

At epoch 18, CNN achieves 85.3% F1 score. ResNet-50 attains a F1 score of over 70% at epoch 10.

Baseline reaches 83.9% F1 score at epoch 24. As shown in Figure 2, CNN reaches 72.5% F1 score at epoch 14.

3 Discussion

We note that the training was stable for all models.